CXR-BMD — change versions & re-deploy

Deploy path on the server:

```
None
/home/anson/cxrbmd
```

Compose file used in production: `docker-compose.deploy.yml`
External Docker network: `rapier-carpl_docker_backend`

Components

Service (compose)	Container name	Image (current)
`dicom-listener`	`cxrbmd-dicom-listener`	`dicom-listener:latest`
`ai-worker`	`cxrbmd-ai-worker`	`ai-worker:latest`
`dicom-sender`	`cxrbmd-dicom-sender`	`dicom-sender:latest`
`redis`	`cxrbmd-redis`	`redis:7-alpine`

Integration notes (unchanged by version bumps unless you edit env):

- Listener AE / port: `AI_ENGINE` / `11112` (CARPL pushes here)
- Sender pushes results to `rapier-orthanc-prefect:4242` AE `CARPL-RECIEVE`
- Shared data under `./data/incoming` and `./data/output`

Algorithm configuration (CARPL Platform registration)

When registering the CXR-BMD algorithm in **CARPL Platform**, paste the following into **Algorithm Configuration**.

```json
{
  "ip": "cxrbmd-dicom-listener",
  "port": "11112",
  "ae_title": "AI_ENGINE",
  "modality": "DX",
  "algo_type": "dicom-push",
  "use_algo_specific_postprocessor": false,
  "private_findings": {
    "creator": "CARPL_AI",
    "creator_tag": "0013,0010",
    "findings_tag": "0013,1001"
  }
}
```

Field	Meaning
`ip`	Docker container name of the AI DICOM listener on `rapier-carpl_docker_backend` network
`port` / `ae_title`	Must match listener env (`11112` / `AI_ENGINE`)
`algo_type`	`dicom-push` — orchestrator C-STOREs studies to the listener (This must be set to `dicom-push` for all the algorithms)

`private_findings`	Tags the postprocessor reads on algorithm output DICOM

How to change a component version

Images are pinned in `docker-compose.deploy.yml` via the `image:` field.

1. Load the new image on the server (from a tar)

Copy the image archive to the server, then:

```Shell
# single image tar
docker load -i /path/to/ai-worker-1.2.3.tar

# or if gzipped
docker load -i /path/to/ai-worker-1.2.3.tar.gz
```

`docker load` prints the loaded image name and tag, for example:

```None
Loaded image: ai-worker:1.2.3
```

Use **that exact** `name:tag` in compose (step 2). Confirm if it is present:

```Shell
docker images | grep -E 'dicom-listener|ai-worker|dicom-sender'
```

2. Point compose at the new tag

Edit `docker-compose.deploy.yml`, for example:

```None
  ai-worker:
```

```
    image: ai-worker:1.2.3          # must match what docker load
reported
```

Do the same for `dicom-listener` / `dicom-sender` as needed. Leave `redis:7-alpine` unless you intentionally upgrade Redis.

3. Redeploy

Continue with **Redeploy** below.

Shell

```
cd /home/anson/cxrbmd

# optional: confirm network exists
docker network ls | grep rapier-carpl_docker_backend

# recreate only what you changed (example: ai-worker)
docker compose -f docker-compose.deploy.yml up -d --force-recreate
ai-worker

# or recreate the whole stack
docker compose -f docker-compose.deploy.yml up -d --force-recreate
```

Verify

Shell

```
cd /home/anson/cxrbmd
docker compose -f docker-compose.deploy.yml ps
docker ps -a | grep -i cxrbmd

# quick logs
docker logs --tail 50 cxrbmd-dicom-listener
docker logs --tail 50 cxrbmd-ai-worker
```

```
docker logs --tail 50 cxrbmd-dicom-sender
```

Confirm the running image ID matches what you expect:

```shell
docker inspect -f '{{.Name}} {{.Image}} {{.Config.Image}}' \
  cxrbmd-dicom-listener cxrbmd-ai-worker cxrbmd-dicom-sender
```

Tips

- Prefer **explicit tags** (`ai-worker:1.2.3`) over `:latest` so rollbacks are clear: set the previous tag in compose and recreate.
- Data under `./data/` is bind-mounted; recreating containers does not wipe studies already on disk.
- No host ports are published; traffic is via the Rapier Docker network and DICOM to the listener AE.